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Objects vs Images for Method of Loci in Mixed Reality

Comparing the efficacy of using 3D objects against 2D images in
the Method of Loci

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No statistical
Significance

Could look into
AI generating
images

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Abstract

The method of Loci is a mnemonic where information recall is improved by visualising spatial locations. Recently, a lot of research has been done on the method of Loci in virtual reality, where participants get to memorise an ordered list of objects by placing virtual representations of the objects in virtual environments. Some studies use 2D images to represent these objects, whilst others use 3D models. There is no consensus on how objects should be represented, but this could have an impact on users memory retention. This study tests if there is a difference in memory retention from using the method of Loci with either 3D models or 2D images. A mixed reality method of Loci application was developed where objects could be represented either in 2D or 3D. Using this application, a between-subject test with 20 participants was performed to test if there was any difference in memory retention between the two modalities. The study did not find a significant difference between the 2D and 3D groups. However, the study had multiple limitations such as a limited number of participants, the limited physical space of the location where the test was performed and the limited number of objects. A future study without the limitations in this study could find that there is a significant difference between the two, which could have a great impact on how future studies in the field are performed.

Keywords

Method of Loci, Mixed Reality, Memory

Sammanfattning

Minnespalats är en minnesregel där saker som ska memoreras kods till platser i en välkänd omgivning. På senaste åren har mycket forskning om minnespalats gjorts i virtual reality, där deltagare får komma ihåg en lista med objekt genom att placera virtuella representationer av objekten i en virtuell omgivning. Vissa studier använder bilder för att representera objekten, medan andra använder 3D modeller. Trots att representationen som väljs kan ha en stor betydelse för minnet finns det ingen konsensus inom forskningen om vilken representation som bör väljas. Den här studien testar om det finns någon skillnad i hur väl folk minns efter att ha använt minnespalats med bilder och 3D modeller. En minnespalatsapplikation skapades för mixed reality där objekten kunde representeras antingen i 2D eller 3D. Med applikationen utfördes ett test där 20 deltagare delades in i två grupper för att se om 2D eller 3D gav bäst förutsättningar för minnesbildning. Ingen signifikant skillnad hittades mellan grupperna. Dock hade studien flera begränsningar, såsom antalet deltagare, platsen det utfördes på, mängden objekt i studien, osv. En framtida studie utan samma begränsningar skulle kunna hitta en signifikant skillnad mellan de 2D och 3D, vilket skulle kunna ha en stor effekt på hur framtida studier inom området utförs.

Nyckelord

Loci-Metoden, Mixed Reality, Minne

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List of acronyms and abbreviations

AR	Augmented Reality
HDRI	High Dynamic Range Imaging
JSON	JavaScript Object Notation
MoL	Method of Loci
MR	Mixed Reality
VMoL	Virtual Method of Loci
VR	Virtual Reality
XR	Extended Reality

Chapter 1

Introduction

1.1 Background

Method of Loci (MoL) is a mnemonic device for memorising ordered lists of items and has been used since ancient Greece. It works by encoding items to be remembered to spatial locations, which the human brain is particularly apt at recalling [1]. It has long been regarded as an efficient mnemonic, capable of improving memory recall [1]. The method itself is rather simple:

Imagine yourself in a familiar location, such as your home. Imagine walking through your home and placing the items in different locations along a path. To recall the items later, imagine walking the same path and seeing/picking up the items as you walk along.

The introduction of computers and interactive software such as video games has allowed **MoL** to take on new forms. **Virtual Method of Loci (VMoL)** refers to virtual versions of **MoL**, where users of the mnemonic navigate and place items in a virtual world rather than using their imagination. This has clear benefits for researching the method, as it allows the researcher more control over the experience, and gives full visibility of the participant's actions. Early **VMoLs** tended to use flat screen displays, but recent technological developments have broadened the research field to using technologies like **Virtual Reality (VR)**, and to a lesser extent **Augmented Reality (AR)** and **Mixed Reality (MR)**, to allow for increased immersion in the environment. Multiple studies have shown that the increased immersion afforded by **VR** is better for **VMoL** than traditional methods [2], [3], [4].

1.2 Problem

When it comes to studies about **VMoLs** in **VR**, some studies represent the objects to be remembered with pictures, whilst others represent them using 3D models. Could there be a significant difference between these two ways of representing the objects? Since the reason to use **VR** is to increase realism, one might assume that representing the objects as 3D models would improve immersion, and thus be a better option for **VMoLs**. This study aims to research whether representing the objects in a **VMoL** as 3D models significantly improves memory retention over using images.

*Does displaying objects as 3D models as opposed to 2D images in a **MR VMoL** improve users' ability to remember an ordered list of items as measured by how many items they remember and how well they remember the order of the items both right after exiting the **VMoL** and after a week has passed.*

The hypothesis is that using 3D models will create more immersion than 2D images since objects that we interact with day to day tend to be three-dimensional. This is hypothesised to lead to greater long-term memory retention.

1.2.1 Original Problem definition

Originally, the idea of the project was to test if familiarity with the space that an **MR VMoL** was performed in increased the memory retention of users. Participants would be split into two groups, one that performed the test in a familiar space such as their home, and one that performed it in a location that they were unfamiliar with. The hypothesis was that participants who performed the **MoL** in a familiar space would have an easier time thinking themselves back to the space and would thus have greater long-term memory retention. The issue that arose was that it would be extremely difficult to ensure that the spaces were equivalent. If one group performs the test in their own homes, different people may have differently sized homes and with different amounts of furniture, creating a lot of noise making it difficult to know if trends in the data came from having previous knowledge with the location or just the locations being different. Thus, it was decided that it would be better to be in the same location, and instead vary the representation of the objects that participants were shown, as that would theoretically produce data with less disturbance.

1.3 Purpose

The thesis aims to expand the scientific knowledge base on how different representations of objects affect a subject's ability to memorise items in a **VMoL**, by examining the difference between 2D images and 3D models in a **VMoL**. Papers posted on the subject seemingly do not put much thought into whether to use images or 3D models, so the result of the study could potentially have a large effect on how future studies on the subject are performed, given that one of the representations shows a clear boost in memory retention. Whilst the study is performed in **MR**, the findings would likely generalise to other head-mounted display technologies like **VR** and headset-based **AR** as well. However, specific studies should be performed to certify that the findings also apply to other modalities.

Improving our memory capacity through technology has many clear benefits, and continued research into learning in **Extended Reality (XR)** can have a lot of impact on a societal level. It could help the elderly retain their memories and brain function for a longer time. It could revolutionize education by letting students learn and study in VR. The research field has many potential benefits both on an individual level and for society as a whole, and this study hopes to serve as one small step forward for the field.

1.4 Goals

The goal of this project is to:

1. Goal 1: Create an **MR** application for using the **MoL**. The application should be simple enough that even people without experience in **XR** or video games should be able to use it after a simple explanation.
2. Goal 2: Perform a study using the application where participants are split into a 2D and a 3D group, in order to see which method of displaying the objects leads to better memory recall.

1.5 Research Methodology

To get a general idea of the field and what research has been done previously, a literature study was performed looking at studies on **VMoL** performed in **VR**, **MR** and **AR**. There were very few studies in **MR** and **AR**, but a good amount

of studies had been done in **VR**. Studies that were of most relevance for the project were picked out.

After the literature study, a **VMoL** application for **MR** was developed with two modes, one where participants would use 2D images to perform the **MoL** and one where they would use 3D models. A user study was then performed. Participants were split into two groups, each corresponding to one of the modes of the application. They used the application for 20 minutes to remember 20 words in order. They were tested on how well they remembered the words and the order of the words both right after using the application and after about a week had passed. This gave data about how short-term and long-term memory was affected by using either 2D images or 3D models.

1.6 Delimitations

The thesis will only deal with **MR** and the results may not apply to other forms of **VMoL**. It is assumed that the similarities between **VR** and **MR** mean that the results would apply to **VR** as well since the representation of the objects is the same and only the environment changes, but further research would be required to certify that this is the case.

Since the study uses human subjects and the number of participants is limited, it may not generalize to the greater population.

The objects that participants needed to remember were limited to physical objects only. They were not asked to remember words like "time" or "autumn" which would not have an exact physical representation. Including such words could potentially change the results.

Whilst previous studies recommend that the word lists given to participants contain at least 40 words and that longer training times are needed to be used to fully show the potential of a **VMoL**, it was decided that this study would be limited to 20 words and 20 minutes in the **VMoL** to ensure that not too much of participant's time was used [5].

The space that was used was also rather limited. To ensure that all participants could perform the test in the same place, a room at KTH with many unbooked times was chosen as the location for the experiment. Previous research has suggested that the space used in a **VMoL** should be as immersive as possible, but also that it works best when there are unique spots to place items in and there is enough room in the environment to place items with ample space between them [1] [6]. Whilst the environment is highly immersive by using **MR**, the locale used for the experiment was limited to a single room without many unique places for items. This is a factor that also affected the

amount of items that could reasonably be used in the study, as there would not be space to put down too many items.

Chapter 2

Background

2.1 Method of Loci

For millennia, mnemonic devices have been used by humanity to increase our capacity to remember information. Mnemonics are memory strategies that allow you to clump together different pieces of information you wish to recall by relating them to more easily remembered things. **MoL** is a mnemonic which was created in ancient Greece to allow philosophers to remember the lectures that they would hold for their students [1]. Though it was created long ago, it is still considered to be one of the most efficient mnemonics today [1].

The **MoL** is used to remember ordered lists of items by connecting them to spatial locations, which the human memory is well adapted to remember. It is performed by imagining oneself in a familiar place, such as one's own home, where one creates a "path" to walk along. For example, starting in the kitchen, going into the living room, walking into the bedroom and then out again, etc. Whilst walking along this path, one imagines placing the items to be remembered in different locations, for example on the kitchen table, along a windowsill, and on the sofa. To recall all the items and their order, one has to imagine walking the same path again and seeing all the items.

There are some major drawbacks with the **MoL**. Since it requires you to imagine yourself walking through a place, a good visual imagination is required. This can make it difficult to perform for some people who don't have a good visual imagination, and be completely impossible for aphantasiacs, who can't visualise at all. The method can also be difficult to teach to people, and long training sessions are typically required before people can use it, making it impractical for studies. Since the method is performed mentally, it is difficult to research since you can't see what a participant is doing.

2.2 Virtual Method of Loci

The drawbacks of a traditional **MoL** can be mitigated by using a **VMoL**. A **VMoL** is when the **MoL** is performed in a virtual world, such as a video game, instead of being performed in one's imagination. This has several benefits. It is much easier to teach the method since the person being trained can try the method with the help of an application. After having learnt to use the method, they can then apply it on their own if they wish to do so. It is more accessible since it does not require a strong visual imagination. For research purposes, it gives researchers full control and visibility over the experiment, allowing them to make sure that participants use the method correctly, place all participants in the same location, see where participants place the items, how long it takes them to decide on each placement, how they then practice by re-treading the path, etc. Legge *et al.*, showed that using a **VMoL** does not attenuate the performance of the method compared to the traditional protocol of imagining a familiar place, even if the virtual environment is only briefly introduced [7].

2.3 Virtual Reality

VMoLs started being used in the late 90s by using traditional computer screens, but as the years have gone on and technology has advanced, **VMoL** has been adapted to various technologies. One of the more popular technologies used over the past decade is **VR**.

VR is an alternative to traditional screens, where a headset-mounted 3D display is used to create greater immersion. The headset is tracked in space to allow the user's movements to be reflected in the virtual world. The headset is usually paired with either position-tracked controllers or hand tracking, allowing the user's hands to be present in the environment. Depending on how much space is available, the user can move around by simply moving their body, rather than using buttons on a keyboard or a joystick. However, the virtual space is usually larger than the physical space, requiring virtual movement controlled via a joystick on a controller to be mixed with the physical movement. One of the strengths of **VR** is the sense of presence gained through the heightened immersion [8].

Multiple studies have shown that **MoL** in **VR** can be efficient. One study found that the first time subjects used a **VR MoL**, they were able to memorize 20.4% more words than they could by using pen and paper [2]. When the same subjects used the **VMoL** a second time they remembered 22.2% more

than with pen and paper. This indicates not just that **MoL** in **VR** is effective, but also that practice can make users even better at remembering items using a **VR MoL**. One study found that compliance rates for using **MoL** in a **VMoL** was higher in **VR** compared to a flat-screen **VMoL** (97.4% contra 77.5%) [3]. Another study showed moderate improvements using a **VR MoL** compared to using an image-based variant of the **MoL**, and greatly improved results over a control condition without a pre-planned strategy [4].



Figure 2.1: The Quest 3 is a headset that can be used for both **VR** and **MR**. It has a set of cameras on the front to track its location and the location of the controllers.

2.4 Mixed Reality

An alternative technology to **VR** is **MR**, where cameras on a **VR** headset are used to capture the environment around the user to show it on the screens in the headset, creating a passthrough allowing the user to see the real world around them. Virtual objects can then be added by displaying them on the screen along with the real environment, and they can be interacted with using controllers or finger tracking. Rather than using **VR**, **MR** was used for this study due to its relative lack of motion sickness, intuitiveness for moving around, and the fact that interest in **MR** has risen recently with the release of headsets like the Meta Quest 3 and the Apple Vision Pro. A recent study by Huttner *et al.*, comparing memory retention in **VR** and **MR** found that subjects had greater

memory retention in MR than VR [9]. The study was not on MoL, but it shows that MR has potential for use in VMoL. Due to the relative recency of MR, there were quite a lot of studies on VMoL in VR but a distinct lack of studies in MR at the start of the study in this thesis, which meant that the study could help expand the research field by introducing the use of a different type of technology.

2.4.1 The Importance of Immersion

Previous studies have shown the importance of immersion for memory retention in virtual environments. In a study by Huttner, *et al.*, it was found that displaying images along with text to participants in a VMoL was better for long-term memory retention than just showing text [5]. It was recommended that future research studies the effect of adding different representations, such as adding animation to the images. Although 3D models were not specifically mentioned, this paper could be seen as a continuation of the research done by Huttner, *et al.*, since it further explores how objects should be represented in a VMoL to optimize memory retention.

Studies have also shown that it is important for immersion to show a representation of the user's body in VR [10] [11]. Not having the body shown can create dissonance in the experience, and it is recommended that the user's movements are tracked so that a representation of their body and movements can be added to the virtual world. By using MR instead of VR, the user can see their own body through the passthrough, creating a more immersive experience.

Moll *et al.*, found that participants who rated themselves as being more immersed in the experience had better performance in memory recall after using a VR MoL [2]. This shows that improving the immersion should lead to greater results in a VMoL. The study used special shoes from KatVR which allowed users to stand in place and mimic a walking movement by lifting their legs to control their character. The authors recommended that future studies should use a frictionless platform for users to walk on or an omnidirectional treadmill in order to create a more immersive experience where users could walk like in real life. Similarly, Sanchez *et al.*, work has suggested that haptic feedback from the environment and moving through the environment by physically walking enhances presence in a virtual environment[11]. By using MR, this study will allow users to move physically through the environment without the need for frictionless platforms or omnidirectional treadmills since the environment used is real rather than virtual. Using a real environment will

also give haptic feedback, as moving your hands through walls in the real world is impossible. This should create a more immersive experience, and lead to better memory retention.

2.4.2 Mixed Reality vs. Virtual Reality

Using **MR** for a **VMoL** as opposed to **VR** comes with a lot of benefits, but also some serious drawbacks. As discussed, the increased immersion that **MR** gives should lead to greater memory retention than using **VR**. Another benefit is that, by using a real environment rather than a virtual environment less processing is required on the hardware as only the objects themselves need to be rendered. Lower-end hardware such as the quest 2, or even the quest 3 to an extent, could struggle to render a highly detailed scene meaning that sacrifices must be made in the amount of realism and detail that can be displayed. Creating an optimal scene for **MoL** also takes a lot of time, which is not required when using a real location. However, potentially the greatest benefit of a commercial **MoL** application using **MR** might be that users can use an environment that is already familiar to them, such as their home. Traditionally, **MoL** is performed using a familiar location, as users have to imagine themselves being there. Research has shown that a **VMoL** in an unfamiliar environment can rival traditional **MoL**, but using a **VMoL** in an environment that the user is highly familiar with might create even better long-term memory retention, as they will have an easier time recalling the environment and the path afterwards. Whilst this study will not use a location that is highly familiar to all participants, it is something that future research could cover.

Using **MR** also has some potential drawbacks, depending on how it is used. Since a real location is required, there can be problems with using a **MR VMoL** if the location available is not well suited for **MoL**. Previous research has suggested that an optimal environment for **MoL** should have ample space to ensure that users can place items with room between them, and interesting locations to place the items. For research purposes, it is important that the location used for all participants is the same to ensure that differences in the locations used did not contribute to differences between participants. If a space fulfilling those requirements cannot be used, **MoL** will not be fully efficient. However, whilst it may not be fully efficient, it should still be usable. Whilst there are drawbacks to using **MR**, it was decided that it would be used in this study for its many potential benefits.

2.5 3D vs. 2D

Previous studies of **MoL** have represented the objects to be remembered in different ways. Whilst a lot of studies opt for using 2D images, some studies use 3D models instead (specifics studies are described in sections 2.5.1 and 2.5.2). Whilst this may seem like an important decision, there does not seem to be any consensus over which is better and no study has been made in which the two modes were compared. Some studies do not show or describe how their objects were displayed, making it impossible to know if they used images or 3D models. Some early studies used a virtual environment but did not have objects for participants to place, requiring them to imagine the objects instead. Given the importance of immersion as previously discussed, and the fact that objects in the real world tend to be three-dimensional, it could be hypothesized that using 3D models would create a more immersive experience where the objects feel more real, which would then result in better memory retention. Let's take a look at some studies using 2D and some studies using 3D, and look at their respective benefits.

2.5.1 2D

Using images comes with different potential benefits, especially if geared towards consumers of a potential application rather than in research. Firstly, rendering images is extremely efficient, as they only need four vertices and thus only two faces (tris). This is especially important when using lower-end devices with less powerful hardware, like the Quest 2, as using realistic models with high poly counts could tank the performance and make the experience laggy for the user. It's especially important if the amount of items to be remembered is high. The second major benefit is ease of use and set-up. It is much easier to find high-quality free images than it is to find high-quality free 3D models. Whilst it's definitely possible to find enough for a study, it might be difficult to get enough models in a similar art style to have enough to make a commercial application. This leads on to the last benefit, which is customization. If creating a commercial **MoL** application where the user can remember any set of words that they want, the application would either have to have prebuilt 3D models/images for any word the user could conceivably want to remember, or users would need to be able to add their own 3D models/images. Since most people have a smartphone, it could be used to take pictures of things the user wants to remember which would be uploaded to the application. This would allow users to have objects that are

very specific to them. For example, if the word wine is to be remembered, the user could upload an image of their favourite wine to use in the application. It would be much more difficult to do this using 3D models, as most people do not have the ability to create 3D models. Further development in techniques like Gaussian splatting might lead to a future where users could generate their own 3D models easily, but for now, images allow for much easier customisation.

2D images have been used in many studies regarding **MoL**. It is the most common way of displaying the objects, presumably because it is simpler to find and implement images than 3D models. Moll *et al.*, used images and allowed participants to make a selection between 3 images [2]. They found that participants increased their lenient score by 20% when using the **VR MoL**. A study by Huttner *et al.*, also used images along with text to represent their objects to be remembered [5]. They found that participants remembered better than would be expected based on the Ebbinghaus' forgetting curve [12].

2.5.2 3D

3D models have their own distinct advantages over images. Since the items we interact with in day-to-day life are three-dimensional objects, it stands to reason that they would create a more immersive experience, which could lead to better memory retention for users. This, of course, needs to be studied, and is the main theme of this thesis. Another benefit of 3D models is that the user can interact with them more naturally. A sword can be held and swung, a piano can be played, food can be eaten, etc. Since the items are represented three dimensionally, just like real objects, they can more easily be interacted with as though they were real. Using the objects and playing with them in different ways might be beneficial to memorizing them better, even though it is not necessarily a part of **MoL** itself. Another benefit is that a 3D model can be viewed from any angle, whilst a picture can only be viewed from the front (unless it rotates to face the user).

Different studies on **MoL** have used 3D models. Reggente *et al.*, used 3D models in their **VR MoL** study [13]. They tested whether the explicit binding of objects to spatial locations increased subsequent memory recall and found that allowing users to bind objects to locations rather than simply showing the objects in order increased memory recall by 28%. Cho applied **MoL** to language learning using 3D models along with Korean words and found that participants were more motivated and satisfied when using **MoL** in **VR** over flatscreen [14].

2.6 Summary

Studies on **VMoL** use either 3D models or images to represent the objects that participants are supposed to remember. This study will compare the usage of 2D images and 3D models in a **VMoL** to see if one leads to greater memory retention. If one is shown to be better, future research in the field can focus on using that method of visualizing the objects. This study will be done in **MR**. Very little research has been done on **VMoL** in **MR** even though it could have greater immersion than **VR**. By performing this study in **MR** rather than **VR**, the author hopes to expand the research field into new technologies and to explore what the positives and drawbacks are compared to **VR**.

Chapter 3

Method and Implementation

3.1 Overview

A user test had to be conducted to study the effects of using 2D images vs 3D models on memory retention using **MoL**. The test used between-subject design, with one group using 2D images and one group using 3D models. To do this, a custom **MR MoL** application was created. The application would have an ordered list of 20 objects for participants to remember and participants could place the objects freely in the environment around them to create their path. The study itself would have the participants go through a short tutorial to learn the controls, and then use the application for up to 20 minutes to memorise the items as well as possible.

3.2 Hardware/Software used

The Meta Quest 3 was chosen as the headset for the experiment for its coloured passthrough at a reasonable price. The application was developed using Unity (version 2022.3.20f1). Unity was chosen as the engine due to its simplicity in working with **XR**s applications and its high amount of documentation and tutorial videos. Having previous experience with the tool also factored into the decision to use it for this project.

Blender was used for preparing the models and images to be used in the application. Blender is a free open-source program for 3D modelling with good functionality and a large user base, so it was perfect for preparing the models. Having previous experience with the tool also factored into the decision to use it for this project.

3D models were taken from the sites Poliigon and Poly Haven, due to their

large offerings and high quality models and textures. The models for Poly Haven were free, while the ones from Poliigon were paid assets

3.3 Application creation

The application was made in Unity using the Meta XR All-in-One SDK. Since the application would only be run on a Meta headset, using openXR would not come with any substantial benefits. Meta's SDK was thus chosen since it has a tighter integration with the Quest 3.

3.3.1 User interaction

The Quest 3 supports using either controllers or finger tracking as an input method from the user. Both were considered for the project, but it was ultimately decided that controllers would be a better option, since finger tracking and pose recognition produce a degree of uncertainty as the user's intended action may not be received by the application. To ensure that users without experience in XR or games in general would be able to use the application, the interactions needed to be as easy as possible.

The way that the user interacts with the application is by placing and moving objects. Other interactions, such as opening a menu to see the entire list, were considered but ultimately scrapped to ensure that the application would remain simple to use. It was imperative that the interaction would be simple and not get in the way of the user performing their task, so the ways that the user would interact with the application were heavily considered to make them as simple as possible. Ultimately, the interaction ended up using only two buttons. One for placing items, and one for moving items that were already placed. For each object, the user would see a "holographic preview" of the object at the tip of their right controller until they pressed the button to place it down. Once they did, the object would materialize in the same location as the preview had been. If the controller was placed next to an object that had already been placed, the trigger could be pressed and held to move the object. The placing is demonstrated in figure 3.1.

3.3.2 Objects

The 3D models were taken from Poliigon and Poly Haven. Poliigon's models are paid while the ones from Poly Haven are free. This allowed for using high-quality models with realistic textures without the work required to produce

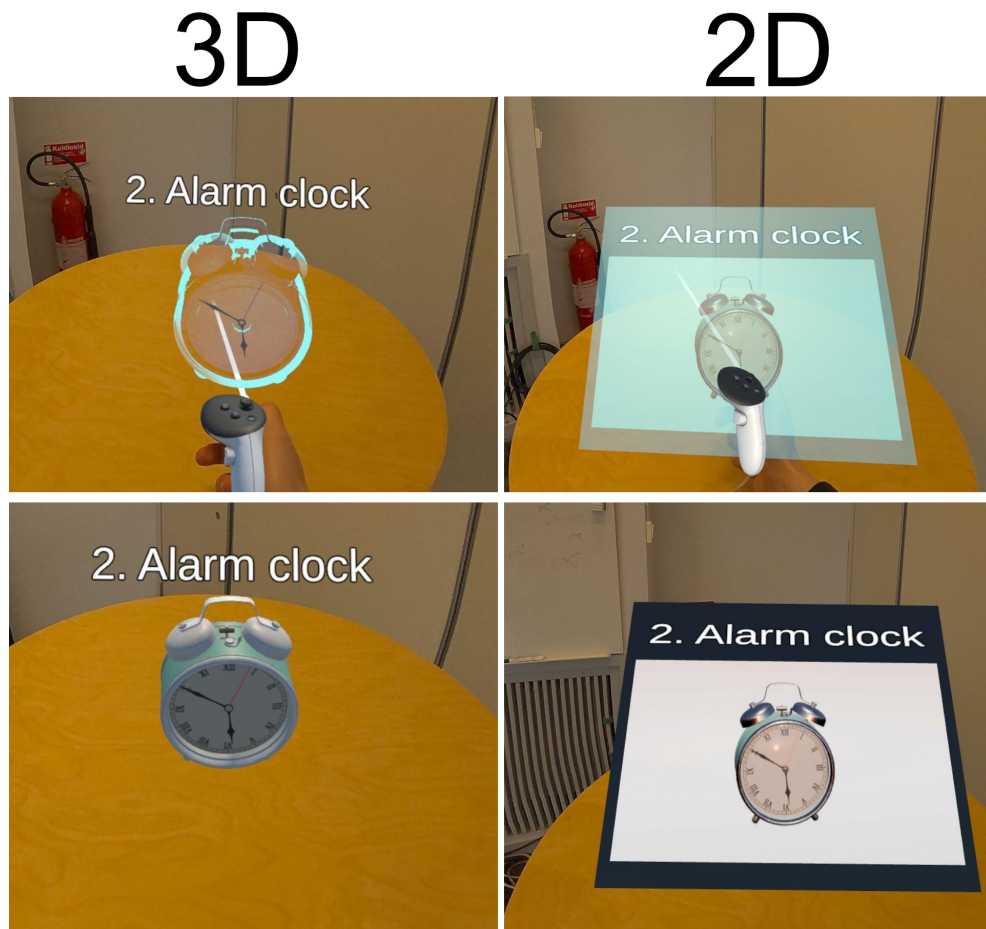


Figure 3.1: Shows how the models and images look before and after placing. The upper row shows the hologram version shown to users before they place the object, and the lower row shows how the objects look when placed.

them from scratch. As discussed in section 2.4.1, increased immersion leads to improved memorization, so realistic models were selected rather than stylised ones, since they are more similar to what we see in day-to-day life and fit better with the real surroundings. A large pool of items was downloaded from which 20 items were selected. The selection included items from different categories, such as food, household objects, fantasy, and tech, to ensure that a wide variety of objects were represented. This was done to ensure that the results would apply to different kinds of items, and not just a single category. Every item had to be something that most people would be familiar with, even if they don't interact with it daily. The full list of objects can be seen in the

appendix. One of the models was altered to reduce its vertex count since the Quest 3 struggled to render the amount of polygons.



Figure 3.2: The 3D models as seen in the application

To create the 2D images, a virtual camera was set up in Blender to capture photos of the 3D models. A neutral white background was also added. For lighting, the combination of a **High Dynamic Range Imaging (HDRI)** from Poliigon and a sun lamp was used. The render engine Eevee was used to render the images in Blender since it produces images more quickly than Cycles, and not using path tracing for the images meant that the 2D and 3D representations would be more similar since Unity uses rasterisation to render the 3D models as well. The images were rendered with the standard setting at 1080p. The scene setup can be seen in 3.4.

The models were exported from Blender as FBX files and imported into Unity. The pictures were saved as PNGs and imported to Unity. For both the images and the models, a text was added on top. If there was no text and users had to go by just the objects, they might come up with different names for the items. For example, if the users were shown orchids, one might remember them as "orchids", one as "flowers" and one as "a plant". This would make it difficult to know if they were remembering the correct items or not. Appending a text with the object's name was the simplest way to ensure that all users

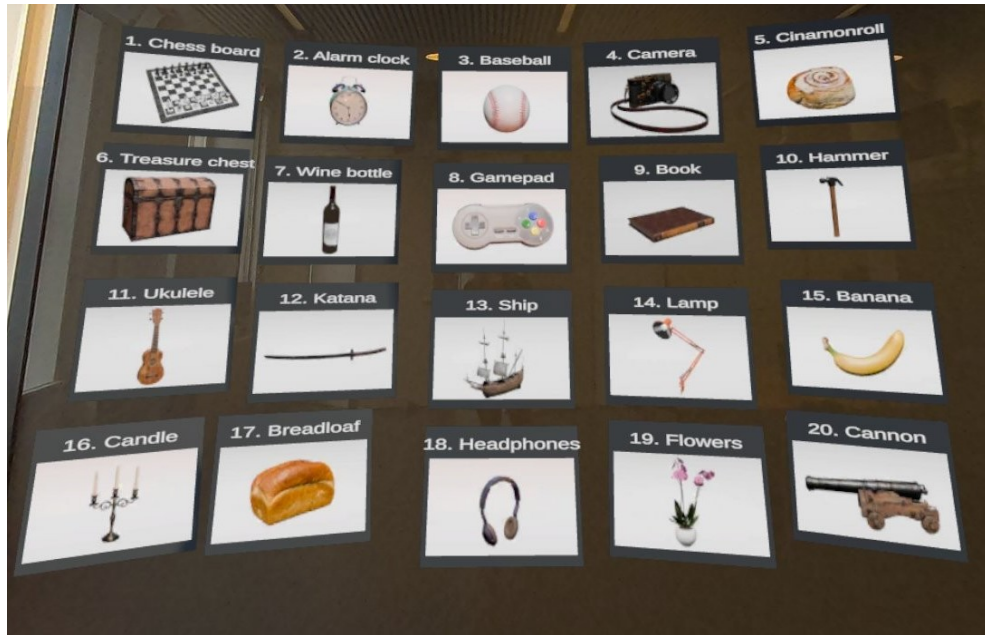


Figure 3.3: The 2D images as seen in the application

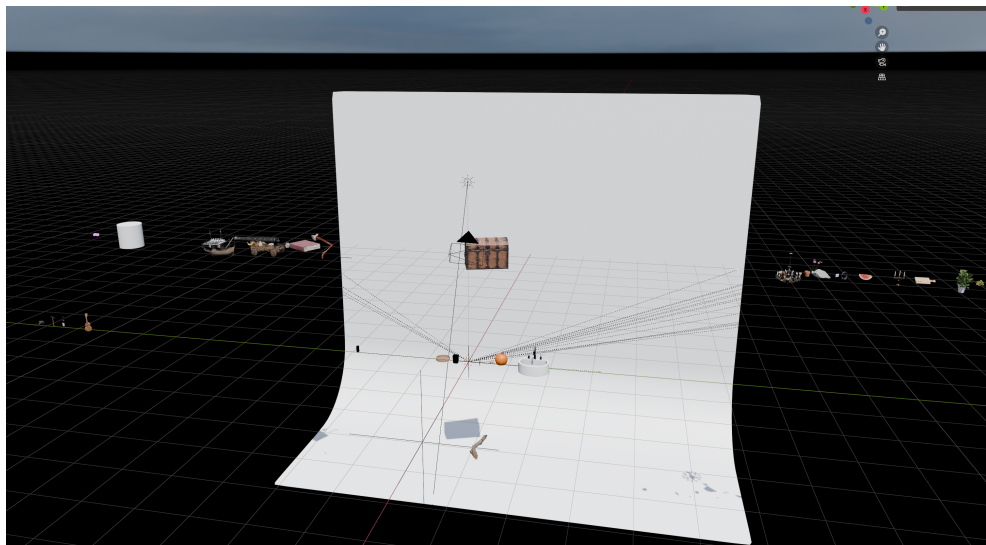


Figure 3.4: The scene used for taking photos in Blender. Currently, the chest is being photographed. Items that were not included in the final test are also seen.

tried to remember the items using the same words. For the pictures, a dark background was added around the text and the image, whilst the 3D models were separate from the text.

A 3D model can be seen from 360 degrees, while images can only be seen from the front. This could potentially give the 3D models an unfair advantage as they would always be visible. To remove the advantage, the images were made to always face the user, and the text over the 3D models was also made to face the user.

The order of the items was randomized, but every user was given the same seed and thus the same order. This was done to eliminate different orders as a variable affecting the outcome. If participants had different orders, some might have had an order that was easier to remember. For example, if similar items were clustered together it might be easier to remember them. Thus, it was decided that the items would be randomized once and shown in that order for every participant.

To ensure that half the participants were shown 3D models and half of them were shown images, the application was programmed to show 3D models every other time it was booted and images every other time by saving a file with information about which representation it had used last time and using the opposite one.

3.3.3 Data gathering

To capture information about how the participants used the application, a script was made to save data such as the location of each object, when each object was placed and how long the user used the application into a **JavaScript Object Notation (JSON)** file.

3.3.4 Tutorial

A short tutorial was added to the application, where participants would get to place down 3 objects and try to move them around to familiarise themselves with the controls before starting the actual test. To ensure that the items in the tutorial wouldn't be mixed up with the ones in the actual test, the objects in the tutorial were simple white geometric shapes. A sphere, a cube and a cylinder. Due to the items being so simplistic and dissimilar from the items used in the study, it was hypothesised that participants would not mix up the items between the tutorial and the test.

3.3.5 Locale

The study was performed in one of the lecture rooms at KTH. Whilst it may have been beneficial for memorisation to use a larger space that subjects were familiar with, such as their own homes, it was important to make sure that every participant performed the test in the same place to ensure that the environment was not a factor that could change the results between participants.

The room used was roughly 6 by 5.5 meters, giving a total of circa 33 square meters of space for participants to use. The room had one wall with two big windows, a screen, and a rack for 3D glasses. One wall with a massive screen, a speaker and a door. One wall with a section for stacked chairs, a smaller screen and a whiteboard. The last wall was a foldable wall that was closed during the tests. (Due to cables being in the way there was a slight opening, but it was almost fully closed.) In the room, there was also a large table where the test author was seated and a small circular table. A diagram showing the layout of the room can be seen in figure 3.5 and photos of the room can be seen in figures 3.6 and 3.7.

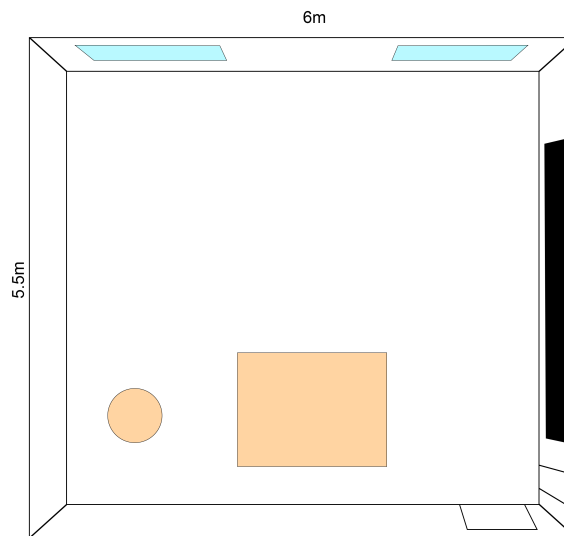


Figure 3.5: A diagram showing the layout of the room. The light blue rectangles are windows, the black rectangles are screens and the beige square and circle are tables



Figure 3.6: A photo of the studio from where the author of the study was sitting. At the bottom, the table where the author sat can be seen. In front is the wall with two windows, to the right is the wall with a massive screen and to the left is the foldable wall.



Figure 3.7: A photo of the studio taken from the opposite side of figure 3.6. Again, the table where the author sat can be seen, but at a greater distance. The circular table is also visible. The wall in the middle has the tucked away chairs and two TV screens, whilst the wall on the right is the foldable wall and the wall on the left has a massive screen

3.4 Data Collection

The data collection was performed using human subjects. Subjects were gathered mostly through classmates from previous courses at KTH. Others were gathered through personal contacts. Though this gives a bias towards computer science students in the data, some participants fall outside this group. This means most participants have a better knowledge of virtual worlds and technology than the general population would.

The computer science students at KTH are mostly male, meaning that the study consisted of more male participants than other genders. Most participants were young, and under 30.

Data was collected through different means. Participants used the application for a maximum of 20 minutes to memorise a series of 20 objects, including their order. They were then given a form in which to write down the objects and their order. This form also contained additional questions, collecting data such as participants' gender, age and level of education. It also contained selected questions from the Witmer and Singer's presence questionnaire [15]. After this, a short interview was conducted with the participants to gain a deeper insight into their thought process and their thoughts on the application. To gain insight into how long-term memories were created, participants would once again get a form to write down the objects and their order after six days had passed. They had two days to answer the form, meaning that all participants performed this step after one week $\pm$ one day.

The subjects were randomly assigned to one of two groups. One group was shown images of the objects whilst the other group was shown 3D models of the same objects. To make sure that the representation didn't differ in other ways than 2D vs 3D, the image group were shown 2D renders of the 3D models taken in Blender. Users were not informed about the fact that two different groups existed to make sure that it would not affect their answers to the questions

3.5 User tests

Participants for the user study were gathered through a variety of means. Most were gained through personal contacts, being either classmates, friends or family to the author, and a few were gathered through KTH with the help of the supervisor for the essay. There were no specific inclusion or exclusion criteria

for the test. Anyone with the ability to perform a test in **MR** was welcome, whether or not they had previous experience with the platform or **MoL**. This decision was made in the hope that the results would better reflect the general populace rather than a selected subgroup.

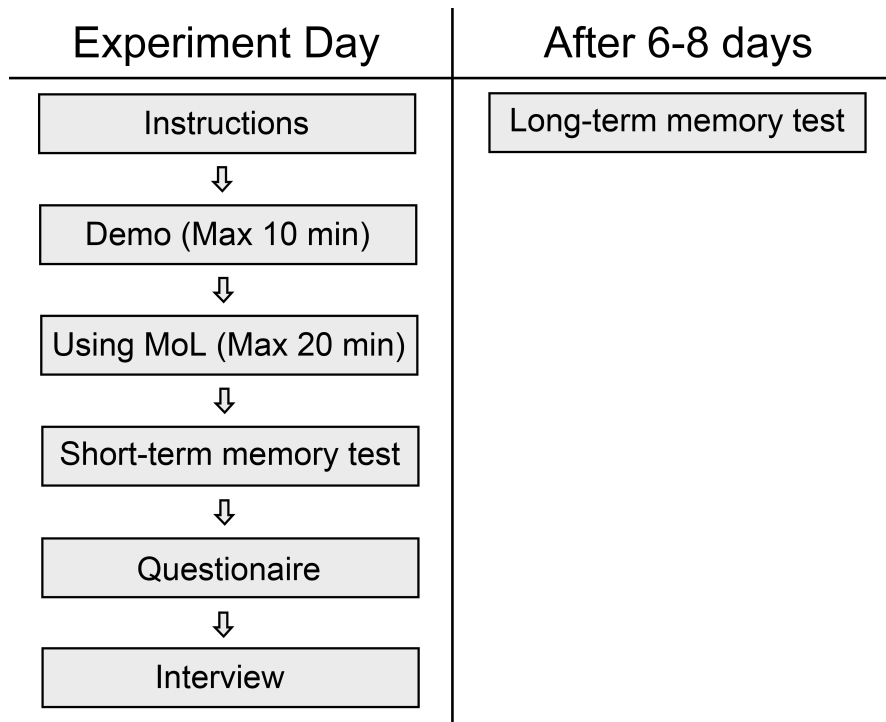


Figure 3.8: A chart showing the parts of the user test and their order.

For each participant, the test was split into 4 different parts, occurring on two different days. On the first day, participants used the application, filled out a form and then a semi-structured interview was conducted. After 6 days, another form was sent out for participants to fill out. They were instructed to fill out the form within two days of receiving it.

First, each participant filled out a consent form and got a basic briefing about what the **MoL** is and how it is performed. They were told that they would perform the **MoL** with the help of an **MR** application. The controls for the application were also explained. Then, they put on the headset and tried the controls in the tutorial. After the tutorial, participants were given 20 minutes to place the 20 items and memorise them and their order. Whilst the participants were using the application, the video from the headset was streamed to a phone so that the test supervisor could inspect their strategies

and help out if there were any issues.

Participants were then given a form to fill out, consisting of some background questions like gender and age, a question where they would write down all the objects in order, and select questions from the Witmer and Singer standardized presence questionnaire. See the appendix for all questions. After the questionnaire was filled out, a semi-structured interview was conducted to gain more insight into participants thought-process about how they used and felt about the application.

After six days, participants were sent a form where they would once again write down all the objects that they could remember and their order. They were instructed to fill out the form within two days of receiving it, meaning that they all performed the follow-up test after 1 week $\pm$ 1 day. The follow-up form was meant to test the long-term memory retention in participants. It was decided that participants would have a total of 3 days to fill out the form, in order to ensure that as many participants as possible would perform the long-term memory test.

Chapter 4

Results and Analysis

4.1 Major results

To analyse the results of both the short-term and long-term memory, lenient and strict scores were calculated. They were proposed by Legge, *et al.*, and have been used in numerous studies [7]. Strict scores are the percentage of objects that a user recalled. The lenient scores are calculated using Levenshtein Distances. These are calculated by seeing how many operations are required to turn a list into a target list. The operations allowed are to insert a word, remove a word and replace a word. The score is then calculated by: $1 - \text{Distance} / \text{MaxDistance}$ with MaxDistance being the maximum amount of operation that could be required to transform any list of that length into the target list. Though the lenient score is a value from 0 to 1, it is presented in percentage.

4.1.1 Short Term Memory

When participants were tested on their memory of the items right after using the application, most recalled all items in the correct order. In the 3D group, all participants remembered the list perfectly, whilst the 2D group had six participants who remembered it perfectly and 4 who remembered the list partially. Figures 4.4 and 4.1 show the results of the 2D group.

A two-tailed t-test was performed to see if there was a statistically significant difference between the groups. The lenient score had a p-value of 0.11 and the strict score had a p-value of 0.11. The aim was for a p-value of 0.05, meaning that the difference between the two groups was not statistically significant.

Table 4.1: Short Term Memory - Lenient Score

Objects Recalled	3D	2D
20	10	7
19	0	1
15	0	2

Table 4.2: Short Term Memory - Strict Score

Strict Score	3D	2D
100%	10	6
90%	0	1
70%	0	1
40%	0	1

4.1.2 Long Term Memory

After a week had passed, most participants had a harder time recalling the items, but 8 participants (four from the 3D group and four from the 2D group) still managed to remember all items in the correct order. Figures 4.1 and 4.2 show a more detailed breakdown of the scores.

A two-tailed t-test was performed to analyse the data. The lenient score had a p-value of 0.25 and the strict score had a p-value of 0.57. The aim was for a p-value of 0.05, so the difference between the two groups was not statistically significant.

Table 4.3: Long Term Memory - Lenient Score

Objects Recalled	3D	2D
20	5	4
19	2	1
18	0	1
17	1	0
16	1	0
14	1	0
13	0	1
10	0	1
8	0	1

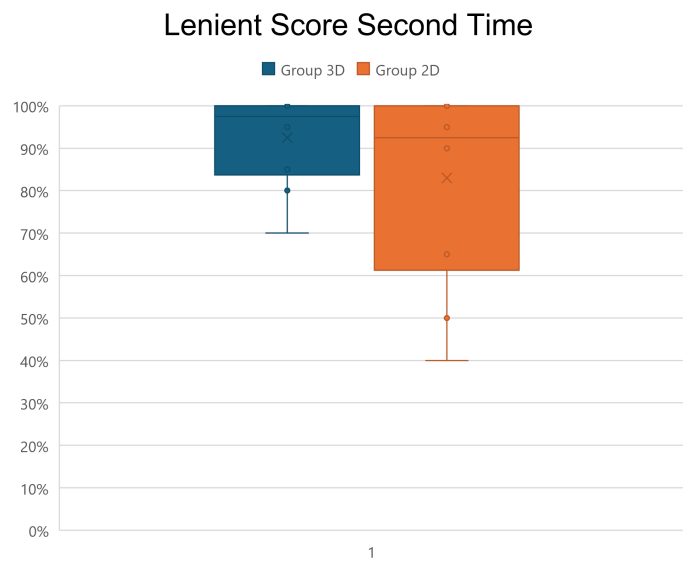


Figure 4.1: A box plot showing the results of the participants. The Y axis shows the percent correct for the lenient score.



Figure 4.2: A box plot showing the results of the participants. The Y axis shows the percent correct for the strict score

Table 4.4: Long Term Memory - Strict Score

Strict Score	3D	2D
100%	4	4
95%	2	1
85%	0	1
80%	1	0
70%	2	0
50%	0	1
45%	0	1
35%	1	0
25%	0	1

4.2 Demography

4.2.1 Age

Most participants were in their twenties, with 15 of the 20 participants falling into this group. In the 3D group, eight participants were in their twenties, one in their 30s and one in their 50s. The 3D group had an average age of 28.8. In the 2D group, two participants were in their teens, seven were in their twenties and one was in their 50s. The average age of the 2D group was 27. The age of the groups is visualised in figure 4.3.

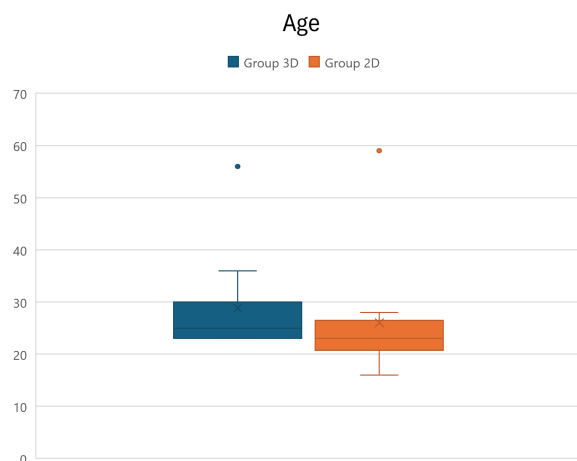


Figure 4.3: Shows the age of participants. Both groups mostly have participants in their 20s, but the average age in the 3D group is slightly higher

4.2.2 Gender

The gender of the participants in the two groups was very similar. In total, 10 women and 10 men participated. In the 3D group, 4 participants were women and 6 were men. In the 2D group, 6 participants were women and 4 were men.

4.2.3 XR experience

Participants in the 3D group reported having more previous experience with headset-based XR than participants in the 2D group. Both groups only had one person without any previous experience, but out of the participants with previous experience, the participants in the 3D group had used VR more. The 3D group had five participants who had developed for headset-based XR, whilst the 2D group only had two. Table 4.5 shows the participant's previous experience in more detail.

Table 4.5: Previous experience with headset-based XR

Experience Level	3D	2D
Developed for headset-based XR	5	2
Use frequently	1	1
Used a bit	3	6
First time	1	1

4.2.4 Education and Occupation

Most participants were students.

In the 3D group, five participants were studying computer science, four were studying media technology and one was a systems engineer. In contrast, the 2D group only had three people studying or working with programming: A research engineer in computer science, a student of computer science and a student of media technology.

4.3 Interviews

The qualitative data from the interviews was analysed using thematic analysis. The interviews were transcribed along with a simplified version where the answers were boiled down to their general concepts instead of listing every

Table 4.6: Area of study/occupation

Area	3D	2D
Computer science student	5	1
Media technology student	4	1
Systems engineer	1	0
Research engineer CS	0	1
Art and media student	0	2
Mathematics student	0	1
Vocational program	0	1
Economy student	0	1
HR	0	1
Unemployed	0	1

Table 4.7: Education level

Education Level	3D	2D
Master's	9	4
Bachelor's	1	2
Other Tertiary Education	0	2
High-school	0	2

detail that was said. Looking through the transcriptions and concepts, ideas that were brought up by multiple participants were found and gathered. These formed the themes that will be presented here. The themes found were visual aid, lack of real-world application, story and association, 3D vs 2D, and controls.

4.3.1 Visual aid

This theme comes from participants stating that they found the visual aid from the application to be very helpful. 18 participants said that they had an easier time using **MoL** with the **MR** application than they would have had if they were to perform it without any visual aid. However, two of those participants stated that they could have used physical objects in place of the **MR** application.

Participant 5 (3D) - "[It's helpful] For many reasons. One is because seeing actual things visually is different from picturing them in your head. And also, I think that when you imagine things in your head, it's easy to focus on one thing and maybe the other goes in the back of your mind and then

you kind of forget about it. It's difficult to split my memory and think about different things when I imagine them, but visually seeing them makes it feel like you have things more under control, and you can go back to the objects whenever you want. But in my brain, if I go back to one thing it's gonna be more effort since I have to imagine it again."

Participant 19 (3D) - *"I think it helped a lot to see the objects, mostly with repetition. I could face the objects and repeat movements and stuff I associated with the objects. Also I thought it was interesting that I made an initial placement of all the objects, but looking back I could quickly see that my method wouldn't work in practice, so I could redo it and learn from my mistakes and try to come up with new ways by moving objects."*

Participant 7 (3D) - *"My ability to visualise is not that strong, so it's very helpful to actually see the things you're placing. After half the time I moved the things a bit which made them feel more like they were there, which also helped"*

4.3.2 Lack of real-world application

Many participants could not see themselves using an application like this in their normal lives. This was mainly due to the fact that they could not find a real-world use for the application. They did not feel that they commonly had to remember ordered lists of items. Some mentioned that whilst they were interested in the concept itself, having to put on a VR headset would be too much of a bother. Multiple participants did mention that it could be useful in a school setting, but not in their current lives.

Participant 19 (3D) - *"I don't know how often I need to memorize a list of items. It's not something I have a need for"*

Participant 16 (2D) - *"I probably wouldn't use this type of application. It's too complicated to set up, takes too much space, and caused a bit of nausea for me"*

Participant 1 (3D) - *"Maybe I'd use it if there was something I would desperately need to remember, but I think it was best as a tutorial for MoL. You don't want to have to pull out a headset every time you want to remember something. Maybe I'd use it sometimes, but I think it works best as a tutorial"*

4.3.3 Stories and association

This theme is about creating stories or using associations to remember the objects. Eleven participants expressed that they utilised stories or associations

to help them remember the words. Some created an entire story to remember all the items, some performed actions with the objects to help them remember, and some only had associations for a few objects.

Participant 2 (2D) - *"At the beginning of the task, I was trying to relate the object to something in real life, like I placed all the electronic devices on the shelf, but then I realised I have to remember the order, so I made up a story for that part. So I lit up the candle and I was in an illusional world where I saw bread listening to music and flowers in the canon, so that makes it easier to remember. So I made a little story with the items, and when I walk through them I remember the story"*

Participant 7 (3D) - *"I created more than just a physical placement. I created a story, or story-ish you know. Nearly every item I had an action connected to it. I didn't just see the items, I used all the items. It felt to me like it helped. I guess the only one I saw was a ship but then I made it into a big deal that it was floating in the air."*

Participant 15 (3D) - *"I used some techniques. First I placed the objects in categories and made a story to connect them. First I was playing chessboard overnight and I forgot to set an alarm, and next morning I was late for going to baseball practice. It's easier to connect and make a story by categorising and placing them in the same place"*

4.3.4 Categorisation

Five of the participants stated that they placed the items in different categories to help them memorize them. For example, putting the treasure chest, ship, and wine together to create a pirate corner, putting food together, etc.

Participant 12 (2D) - *"First I thought I'd place them based on where they would make sense to have, but it's a fairly bare room, so I ended up placing loads of items on tables. So, how do I understand where I've placed the things when everything is on tables? So then I thought I'd put them in categories. But that kind of made me put a lot of things in the same place. Like you had the chest and then a wine bottle which is kind of piratey. Then there was a ship so I also put it there and a candle which I also put there. I tried having the gamepad and headphones together for tech stuff. The banana and the bread together. I had them together in groups, but I had to remember the order so I had to remember going from one category to another and then back to the first one, so I had to put extra effort into remembering the order."*

Participant 5 (3D) - *"I created areas for different categories of objects. For example I had the pirate corner. I think it was categorising objects and making*

stories. For the chessboard I imagined a chessplayer and they need food so that's why I put food by the chessboard. And for the alarm clock, in chess there's like a little clock you had to press so I associated things like that. And I had a music corner, so I was categorising and trying to create a narrative."

4.3.5 3D vs 2D

One theme that came up from multiple participants was the idea of 3D vs 2D. Three participants in the 2D group expressed that they thought the application would be better if 3D models were used rather than images. A participant in the 2D group stated that 3D models would be nice for increased realism, but that 2D models might be better for memorisation. Another participant in the 2D group said that they liked the use of images since it reminded them of study cards (though they did not specifically mention 3D). The participants were not guided by the questions to speak about 3D vs 2D specifically, so it is interesting that so many brought it up.

Participant 8 (2D) - *"I would say that it's better if it's proper 3D models of the objects instead of just cards"*

Participant 2 (2D) - *"I think it kind of works for now. They [the images] do the job pretty well. But maybe if they were 3D objects and you could scale them, like the ship, maybe someone could imagine it as a huge one but I imagined it as a small one like a toy. So maybe you could make them into 3D and the user could decide the scale that would be really interesting."*

Participant 12 (2D) - *"Using 3D objects might have made it more realistic, but for remembering I think this [2D] might be better."*

4.3.6 Colours

Three participants in the 2D group expressed that the colours of the representations could be changed. In the 2D version, the images all had white backgrounds and a grey square around them, which made them fairly monotone. Having the colours be specific to the items was suggested.

Participant 20 (2D) - *"There are specific colours that change how we perceive things. The white background made you focus more on the image and the text. Maybe having specific colours for each object, since it allows you to see and remember the colours."*

Participant 6 (2D) - *"Having stronger colours might make it easier to remember. You could colour-coordinate it as well."*

4.3.7 Controls

This theme is about controlling the application. Most participants stated that they felt the controls were easy to use and did not have any issues with them. However, 4 participants stated that they would have liked to be able to rotate the objects using the stick before placing them. In the current implementation, objects will rotate around the y-axis facing the controller until placed. This means that there is no way to rotate freely unless it is placed down and then grabbed. This was a feature which some participants wanted to have.

Participant 11 (2D) - *"Once you explained how to grab and move the objects it was quite straight-forward. The learning curve was almost non-existent."*

Participant 3 (3D) - *"I'm not sure how much you could rotate the items before placing them. You could rotate around the Y-axis. So if you wanted to rotate something in another axis you would have to place it and then rotate it. So that might be the one thing [to improve], letting you rotate things before you place them."*

Chapter 5

Discussion

The purpose of this study was to compare the memory retention of using 2D images and 3D models in a **VMoL**. The research question was as follows: *Does displaying objects as 3D models as opposed to 2D images in a **MR VMoL** improve users' ability to remember an ordered list of items as measured by how many items they remember and how well they remember the order of the items both right after exiting the **VMoL** and after a week has passed.* It was hypothesised that 3D objects would increase the immersion experienced by participants since they have a greater resemblance to objects that we see in the real world and that this would lead to greater memory retention. The results did not show a statistically significant difference between the two groups, meaning that no improvement was found.

5.1 What did we learn?

5.1.1 3D vs. 2D

Though not statistically significant, the data showed a tendency towards 3D being better for memory retention. The 3D group had greater results in both the short-term and long-term memory tests. During the interviews, multiple participants in the 2D group also stated that they thought using 3D models would improve the application even though they were not specifically asked about 3D vs 2D. This indicates that future studies in the field could have greater results by using 3D models rather than simple images. Naturally, this needs to be confirmed by a statistically significant study, but the signs still point towards 3D being more efficient. This also aligns with previous research showing that greater immersion leads to better memory retention as discussed in section

2.4.1.

Though it might take some more time to set up, more future studies should consider using 3D objects instead of images. Using 3D models for a study adds a bit of complexity, as high-quality 3D models are harder to come by than images. However, it is not that much of an issue nowadays since multiple websites offer free 3D models that can be used in any project. By using these rather than simple images, immersion can be increased which is likely to lead to better results in the studies. Given the idea that greater immersion leads to better memory retention, it is likely that the style of the 3D models should align with the style of the environment, as having objects that don't match the environment would not be coherent with how we expect the real world to work. Future work could look into whether different art styles such as realistic vs. low poly affect memory retention.

5.1.2 Mixed Reality

As this study used **MR** instead of **VR**, there are some interesting insights to be gained for future research. Firstly, given that multiple participants had full scores even after a week passed, it shows that **MoL** is also effective in **MR**. However, if future researchers decide to use **MR** for researching **MoL**, it is important to ensure that a large space with varied locations is available. For example, having a larger apartment where the test can be done, instead of just a single room, gives more room for participants to move around and lets them place items with more distance between them. Having an apartment rather than a fairly unfurnished room also creates more interesting and varied locations to place items. Placing some items in the kitchen, some in the living room, some in the bedroom, etc. allows participants to segregate the items, which might make them easier to remember. When deciding between making a study in **MR** or **VR**, the available space allocated to the test should be taken into consideration. If a large, interesting and varied location is available, using **MR** may be beneficial due to its greater immersion and lack of virtual controller-based movement. However, performing the research in **VR** would be better if a large space cannot be used.

5.1.2.1 Controls

One of the major benefits that was gained from using **MR** in the study is the simplification of the controls and interactions. Normally, studies on **VMoL** need to have fairly lengthy introductions to ensure that participants learn to move around the virtual environment. For example, a study by Moll, *et al.*,

gave participants up to 10 minutes to get to grips with **VR** and explore the environment. By contrast, the participants of this study only needed about a minute to try out the controls before continuing to the test itself. This is mainly because all movement was performed by physically walking around the room rather than having virtual movement. This simplified the controls to using only two buttons, allowing participants to become comfortable using the application extremely quickly. This removes bias against people with motion sickness and difficulty with technology, as it levels the playing field for all participants. This is a clear benefit of using **MR** over **VR** since movement can be done entirely physically, leading to a much simpler experience for the user.

5.2 Applications

Though this study was inconclusive, the field has many applications in the real world. Memorisation is an important part of our daily lives. Using technologies like **VR** and **MR** along with techniques like **MoL** can have positive effects on many different areas. One area where an application like the one presented in this study could be highly beneficial is for improving the memories of people suffering from dementia, other neurodegenerative diseases or loss of memory from old age. A study by Strong demonstrated that VR opens up many new pathways for interventions for those suffering from dementia, but more research is required [16]. Using the application created for this study, people with dementia could combat their memory loss by training in the application, and practising information that is important to them that they do not wish to forget. Of course, this would have to be studied with this group in order to see if **MoL** would actually help for this group, preferably with the help of neuroscientists and psychologists.

Another area where a **MoL** application could be helpful is creating new tools for the school system to improve the ways that students learn new information. As demands on learning keep growing and we require more and more information for complex jobs, tools for facilitating quicker and simpler ways to acquire information are likely to become more sought after. Teaching kids better ways of learning information will help them throughout their education and career. With **MR** and **VR**, it is possible to teach even young kids new ways of learning in engaging and interesting ways. By teaching them **MoL** early, they could use it in their studies to remember both ordered and unordered things. For example, learning the order of events in history, or remembering the order of what they are going to say in a presentation. Another potential area where this could be used is in language learning. By putting

together objects with their names in a target language, users can use the virtual environment to practice a language with the help of **MoL**. In a study by Cho, **MoL** was used for learning Korean and indicated that **VR** was more efficient than flatscreen for learning [14].

5.3 Demography

One interesting observation from the data is that all participants in the 3D group were either studying or working with programming. A good working memory tends to be very useful for programmers since a lot of variable names have to be kept in memory whilst writing code. This may explain why every participant in the 3D group was able to get a perfect score on the short-term memory test.

5.4 Ethics

To ensure the privacy of the participants, all data has been anonymized. Participants filled in a consent form before the study and verbal consent was given before the recording of the interviews.

Whilst the study's true purpose was hidden from participants, they were told everything they would go through before starting the test.

When performing studies in **XR**, there is a risk that participants will experience motion sickness. To mitigate this, **MR** was chosen instead of **VR**. This allowed for the movement to be controlled by moving around in reality, giving a 1:1 experience between the visuals and motion. Due to this, only one participant stated that they had experienced motion sickness during the study.

5.5 Limitations

The biggest limiting factor of the study was the space that it was performed in. The room was only 6 by 5.5 meters and did not have many memorable spots to place items. As discussed in section 1.6, previous studies done in **VR** have recommended having a large space and memorable locations to maximize the efficiency of **MoL**. This lowered the potential of using **MoL** in the space.

Due to the lack of space, only 20 items were used in the test. Preferably, the minimum of 40 items that Ross and Lawrance suggested ?? would have been used, and more time allocated to learning those items. However, given the spatial constraints, it would have been impossible to make a sensible path

with that amount of items. This meant that 16 of the participants got a full score on the short-term test and that 8 participants managed to recall all items in the correct order on the long-term test. This shows that the number of items was too small. Participants getting the full score essentially means that no comparisons can be drawn between them. One participant getting the full score might have been able to memorize 25 items, whilst another would have been able to memorize 35 in the same amount of time. Since there were not enough items for them to show their full capacity, it can essentially be considered lost data. If a bigger space had been available and more items had been used, it is plausible that the results would have been statistically significant.

Another limitation is the number of participants. Due to this being a master thesis rather than a larger study, only 20 participants could be gathered. This meant that each group only had 10 participants. The more participants that a study has, the better. A low amount of participants means that it is difficult to draw any real conclusions from the data, and getting statistical significance is more difficult. This could perhaps have been mitigated by using within-subject design rather than between-subject design, since that would allow each participant to be part of either group. It would also be interesting to get feedback in the interviews from participants who have used both versions of the application. The issue is that this would have required significantly more time from each participant. In the case of within-subject design, it would be preferable to have the subjects the 2D and 3D tests separately to ensure that they wouldn't mix up the lists. Given that the long-term memory test was a week after using the application, the 2D and 3D tests would have to be separated by at least a week. Requiring participants to use the application twice, at least one week apart, would have likely led to far fewer participants and was thus not viable for this study.

5.6 Potential Biases

Some potential biases might have had an effect on the results. It could be argued that the fact that the 2D images rotated in order to always face the user makes them more memorable since they have movement. Of course, the text over the 3D models also moves to face the participant, but the model itself does not move. This is a discrepancy between how the objects are displayed, but the author would argue that it is better than to have the 3D models visible from all angles whilst the images would only be visible from certain angles.

The images had a white background, whilst the 3D models had no background. This creates greater contrasts for the 2D images, making

them stick out more. On the other hand, the 3D models fit better into the environment. The way that the images and 3D models were displayed was chosen to reflect how other studies in the field tend to display their objects with the respective method. However, it could be argued that the differences are too great and that this creates a bias towards one of the modalities. One way of fixing this would be to have the images use alpha and not have a background, or to add a white square behind the 3D models, which moves based on the participant's location.

Participants were randomly assigned to the two groups. This led to the 3D group having a higher level of education and more experience with XR on average. Whilst the controls and experience were simplified in order to make it as easy as possible to make the XR experience less relevant, it could still have affected the results. The amount of participants who used XR for the first time was the same between the groups, but of the participants of the 3D group had used XR more. Given the small amount of participants, it might have been optimal to split the groups in a way to ensure that they would have an equal amount of XR experience, rather than splitting them randomly.

The general memory of participants was not tested. It might be that participants in one group had better general memory. It would have been optimal to test participants' memory with a general test before performing the **MoL** test. To ensure that enough participants could participate, the test needed not to take too long and thus no general test was performed.

5.7 Future work

5.7.1 Larger study

Whilst this study had flaws in using a small space for the study, few items and few participants, a future study could potentially show a statistically significant difference by removing these limitations. This could be done either in VR or MR, depending on whether a large and varied physical space can be used or not. By increasing the amount of space available in the study, participants will have more space to create their paths and more memorable locations to place the items. Having more space would also make it reasonable to have more items for participants to place, making it plausible to use a minimum of 40 items as suggested by Ross and Lawrence [17].

5.7.2 Interaction, animation and sound

Adding interactions with animations and sounds to the objects could increase the level of immersion further. Taking some examples from the items in this study, allowing participants to play the ukulele by making sounds when users put their hand by the strings, swing the katana to make swosh sounds, light the canon to make it shoot out a canon ball, etc. could create a deeper connection to the items and help participants memorise them further. There is a risk that this would make participants rely on the actions with the objects rather than to use **MoL** itself, but if they are instructed to create a path with the objects and then practice by repeatedly walking down that path and interacting with the items in order it would still follow **MoL**. Having one group with interactive items and one group using items without interactivity would allow for comparisons between the two groups.

5.7.3 Mixed Reality vs Virtual Reality

Whilst this study was set in **MR**, it did not have a group using **MoL** in **VR**, and can thus not draw any real conclusions about whether the increased realism of **MR** has any real benefits over using **VR**. A study where a real environment is used for **MR** and a virtual replica of the same environment is used for **VR** could make a deeper investigation into whether there are any real benefits to using **MR**. **VR** naturally benefits from being able to use an environment fully constructed to be optimal for **MoL**, but comparing the technologies themselves when the environment is equal could show benefits to **MR** as well.

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Appendix A

Interview questions

- Do you have previous experience with the method of Loci, and if so, how have you used it?
- Do you think that the application made it easier to perform the method of loci than if you had done it on your own?
- Do you feel like it would be easier to perform the method of Loci on your own after having used the application?
- Would you use an application like this in your own life, and if so, for what purpose?
- Were there any parts of the application that felt difficult/unintuitive to use?
- Did you use any other techniques to memorise the items?
- Make more specific questions i.e. could the objects be improved?
- Could the method of Placing be improved?

Appendix B

Witmer and Singer presence questionnaire

Answers to the presence questionnaire. Min 1 Max 5. The number presented is the average of either group

- How much were you able to control events?
 - 2D: 4.8
 - 3D: 4.6
- How responsive was the environment to actions that you initiated (or performed)?
 - 2D: 4.6
 - 3D: 4.3
- How much did the visual aspects of the environment involve you?
 - 2D: 4.2
 - 3D: 3.9
- How closely were you able to examine objects?
 - 2D: 4
 - 3D: 4.4
- How compelling was your sense of objects moving through space?
 - 2D: 4.6

- 3D: 4.4
- How well could you examine objects from multiple viewpoints?
 - 2D: 3.8
 - 3D: 4.5
- How involved were you in the virtual environment experience?
 - 2D: 4.4
 - 3D: 4.5
- How quickly did you adjust to the virtual environment experience?
 - 2D: 4.3
 - 3D: 4.8
- How proficient in moving and interacting with the virtual environment did you feel at the end of the experience?
 - 2D: 4.8
 - 3D: 4.7
- How much delay did you experience between your actions and expected outcomes?
 - 2D: 1.6
 - 3D: 1.7
- How much did the visual display quality interfere or distract you from performing assigned tasks or required activities?
 - 2D: 1.5
 - 3D: 1.6
- How much did the control devices interfere with the performance of assigned tasks or with other activities?
 - 2D: 1.4
 - 3D: 1.5

